

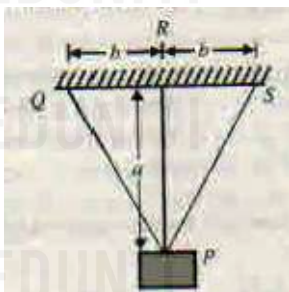
SUBJECTIVE TYPE

- Calculate the gravitational field intensity at the centre of the base of a hollow hemispherical shell of mass M and radius R . (Assume the base of hemisphere to be open)
- Calculate the gravitational field intensity and potential at the centre of the base of a solid hemisphere of mass M and radius R .
- Two point objects, each of mass m , are connected by a massless rope of length l . The objects are held near the surface of Earth, so that the rope is vertical and one object is hanging below the other. Then the objects are released. If $l \ll R$, show that the tension in the rope is $T = \frac{GMm}{R^3}$ where M is the mass of the Earth and R is its radius.

4. A small satellite revolves around a heavy planet in a circular orbit. At certain point in its orbit, a sharp impulse acts on it and instantaneously increases its kinetic energy to k (< 2) times without change in its direction of motion. Show that in its subsequent motion, the ratio of its maximum and minimum distances from the planet is $\frac{k}{2-k}$, assuming the mass of the satellite to be negligibly small as compared to that of the planet.

- A body is launched from the Earth's surface at an angle $\alpha = 30^\circ$ to the horizontal at a speed $v_0 = \sqrt{1.5 GM/R}$. Neglecting air resistance and Earth's rotation, find
 - the height to which the body will rise.
 - the radius of curvature of trajectory at its top point.

- Three elastic wires, PQ, PR and PS support a body P of mass M , as shown in fig. The wires are of the same material and cross-sectional area. The middle wire is vertical. Find the loads carried by each wire.



PARAGRAPH TYPE

Paragraph for Question Nos. 7 to 9

A non-homogeneous sphere of radius R has the following density variation

$$\begin{aligned} \rho &= \rho_0 & \text{for } r &\leq R/3 \\ \rho &= \frac{\rho_0}{2} & \text{for } \frac{R}{3} < r \leq \frac{3R}{4} \text{ and} \\ \rho &= \frac{\rho_0}{8} & \text{for } \frac{3R}{4} < r \leq R \end{aligned}$$

- The gravitational field due to sphere at $r = R/4$ is
 - $\frac{\pi GR\rho_0}{3}$
 - $\frac{2\pi GR\rho_0}{3}$
 - $\frac{2\pi GR\rho_0}{9}$
 - $\frac{\pi GR\rho_0}{16}$
- The gravitational field due to sphere at $r = R/2$ is
 - $\frac{43\pi GR\rho_0}{50}$
 - $\frac{23\pi GR\rho_0}{100}$
 - $\frac{18\pi GR\rho_0}{25}$
 - $\frac{35\pi GR\rho_0}{81}$
- The gravitational field due to sphere at $r = 5R/6$ is
 - $0.98\pi GR\rho_0$
 - $0.48\pi GR\rho_0$
 - $0.23\pi GR\rho_0$
 - $0.52\pi GR\rho_0$

Paragraph for Question Nos. 10 to 12

It can be assumed that orbits of earth and mars are nearly circular around the sun. It is proposed to launch an artificial planet around the sun such that its apogee is at the orbit of mars, while its perigee is at the orbit of earth. Let T_e and T_m be periods of revolution of earth and mars. Further the variables are assigned the meanings as follows.

M_e, M_m = Masses of earth and mars
 L_e, L_m = Angular momenta of earth and mars (each about the sun)
 a_e, a_m = Semi-major axes of earth and mars orbits
 m = Mass of the artificial planet
 E_e, E_m = Total energies of earth and mars

- Time period of revolution of the artificial planet about the Sun will be (neglect gravitational effects of Earth and Mars)
 - $\frac{T_e + T_m}{2}$
 - $\sqrt{T_e T_m}$
 - $\frac{2T_e T_m}{T_e + T_m}$
 - $\left(\frac{T_m^{2/3} + T_e^{2/3}}{2} \right)^{3/2}$
- The total energy of artificial planet's orbit will be
 - $\frac{2m}{M_e} \left(\frac{a_e E_e}{a_e + a_m} \right)$
 - $\frac{2m}{M_m} \left(\frac{a_e E_e}{a_e + a_m} \right)$
 - $\frac{2E_e m}{M_m} \left(\frac{a_e + a_m}{a_m} \right)$
 - $\frac{2E_e m}{M_e} \left(\frac{a_e + a_m}{\sqrt{a_e^2 + a_m^2}} \right)$
- Areal velocity of the artificial planet around the sun will be
 - less than that of earth
 - more than that of mars
 - more than that of earth
 - same as that of earth



LEVEL UP - JEE ADVANCED 2022

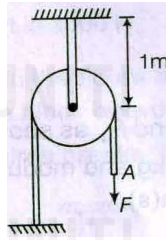
GRAVITATION & SOLIDS

EDUNITI

Paragraph for Question Nos. 13 to 15

The axle of a pulley of mass 1 kg is attached to the end of an elastic string of length 1 m, cross-sectional area 10^{-3} m^2 and Young's modulus $2 \times 10^5 \text{ Nm}^{-2}$, whose other end is fixed to the ceiling. An inextensible rope of negligible mass is placed on the pulley such that its left end is fixed to the ground and its right end

is hanging freely, from the pulley which is at rest in equilibrium. Now, the free end A of the rope is pulled with constant force $F = 10 \text{ N}$. Friction between the rope and the pulley can be neglected. (Given, $g = 10 \text{ ms}^{-2}$)



13. The elongation of the string before applying force is
 - (a) 50 cm
 - (b) 5 cm
 - (c) 0.5 cm
 - (d) 0.05 cm
14. The maximum elongation of the string is
 - (a) 35 cm
 - (b) 30 cm
 - (c) 25 cm
 - (d) 20 cm
15. The maximum displacement of point A after applying F is
 - (a) 70 cm
 - (b) 60 cm
 - (c) 40 cm
 - (d) 50 cm

MORE THAN ONE CORRECT

16. Two satellites S_1 and S_2 are respectively into the orbits of heights $R/4$ and $R/6$ above the surface of earth of radius R . They revolve around the earth in coplanar circular orbits in opposite sense. Which of the following statement(s) is/are correct?
 - (a) The ratio of their time periods is $(15/14)^{3/2}$
 - (b) The ratio of their time periods is $(14/15)^{3/2}$
 - (c) The ratio of their speeds of projection from the surface of earth is $\sqrt{23/15}$
 - (d) The ratio of their speeds of projection from the surface of earth is $\sqrt{21/20}$
17. A double star is a system of two stars of masses m and $2m$, rotating about their centre of mass only under their mutual gravitational attraction. If r is the separation between these two stars, then their time period of rotation about their centre of mass will be proportional to
 - (a) $r^{3/2}$
 - (b) r
 - (c) $m^{1/2}$
 - (d) $m^{-1/2}$

18. A double star consists of two stars having masses M and $2M$. The distance between their centres is equal to r . They revolve under their mutual gravitational interaction. Then, which of the following statements are correct
 - (a) Heavier star revolves in orbit of radius $r/3$
 - (b) Both the stars revolve with the same speed, period of which is equal to $(2\pi/r^3)(2GM^2/3)$
 - (c) Kinetic energy of the lighter star is twice that of the other star.
 - (d) none of these

19. A planet of mass m is moving in an elliptical orbit with the sun of mass M at its focus. If the perihelion and aphelion distances are r_2 and r_1 respectively, choose the correct option(s).
 - (a) Angular momentum of planet relative to sun is

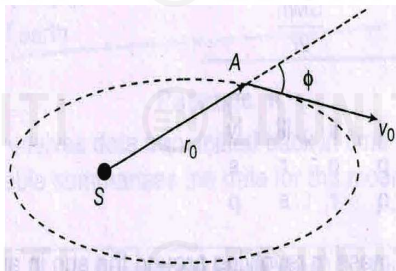
$$L = \sqrt{\frac{2GMm^2 r_1 r_2}{r_1 + r_2}}$$

- (b) Total mechanical energy depends only on the semi-major axis of the orbit
- (c) Total mechanical energy depends on both semi-major and semi-minor axes of the orbit
- (d) $\frac{L^2}{2m} \left(\frac{1}{r_1^2} - \frac{1}{r_2^2} \right) = GMm \left(\frac{1}{r_1} - \frac{1}{r_2} \right)$

20. An earth's satellite is revolving in a circular orbit of radius a with a velocity v_0 . A gun is in the satellite and is aimed directly towards the earth. A bullet is fired from the gun with a muzzle velocity $v_0/2$. Neglecting resistance offered by cosmic dust and recoil of gun during subsequent motion of bullet from the centre of the earth, choose the correct option(s).
 - (a) maximum distance of bullet is $3a$
 - (b) minimum distance of bullet is $3a/4$
 - (c) maximum distance of bullet is $2a$
 - (d) minimum distance of bullet is $2a/3$

21. A planet A at a certain moment of time was at a distance r_0 from the sun and was travelling with a velocity v_0 . The angle between r_0 and v_0 is ϕ . If the mass of sun is M and

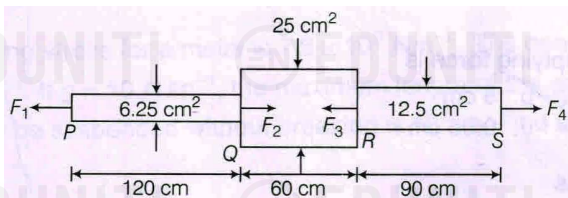
$$K = \frac{r_0^2 v_0^2}{GM}, \text{ then the}$$



- (a) minimum distance of the planet from the sun is $\frac{r_0}{2-K}(1 - \sqrt{1 + K(2-K)\sin^2 \phi})$
- (b) minimum distance of the planet from the sun is $\frac{r_0}{2-K}(1 - \sqrt{1 - K(2-K)\sin^2 \phi})$
- (c) maximum distance of the planet from the sun is $\frac{r_0}{2-K}(1 + \sqrt{1 - K(2-K)\sin^2 \phi})$
- (d) maximum distance of the planet from the sun is $\frac{r_0}{2-K}(1 + \sqrt{1 + K(2-K)\sin^2 \phi})$

22. A member PQRS is subjected to point loads F_1, F_2, F_3 and F_4 as shown in the figure. Given, $F_1 = 4500 \text{ kg}$, $F_3 = 45000 \text{ kg}$, $F_4 = 13000 \text{ kg}$ and modulus of elasticity $= 2.1 \times 10^6 \text{ kg cm}^2$.

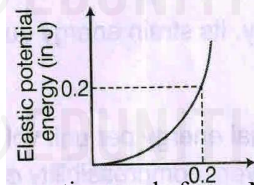
Choose the correct option(s)



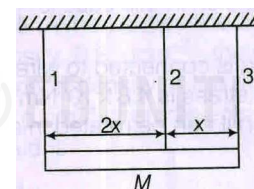
- (a) For equilibrium, $F_2 = 36500 \text{ kg}$
- (b) Total elongation is 0.98 mm
- (c) For equilibrium, $F_2 = 26500 \text{ kg}$
- (d) Total elongation is 0.49 mm

23. Figure shows the graph of elastic potential energy (U) stored versus extension, for a steel wire ($Y = 2 \times 10^{11} \text{ Pa}$) of volume 200 cc . If area of cross-section is A and original length is L , then

- (a) $A = 10^{-4} \text{ m}^2$
- (b) $A = 10^{-3} \text{ m}^2$
- (c) $L = 1.5 \text{ m}$
- (d) $L = 2 \text{ m}$



24. Three vertical wires 1, 2 and 3 are supporting a rod of mass M in horizontal position as shown in the figure. The wires are of equal length and equal cross-sectional area. It is given that $Y_2 = Y_3$ and the rod remains horizontal always. The wires 1 and 3 are attached at the extreme ends of the rod. Choose the correct option(s)



- (a) $T_1 = 2T_3$
- (b) $T_1 = 4T_3/3$
- (c) $T_2 = T_3$
- (d) $T_1 = \frac{2Mg}{5}$